

Saudi Mais Co, for Medical Products Work Instruction

Title (Chapter, #04)
Inspection Procedure for Endobronchial Tube

Doc. #	:	WI-8-2-2-06-1
Rev. #	:	00
Date	:	05.12.2023
Page.#	:	1 / 11

1. Purpose:

Purpose of this procedure to perform inspection during production process of endobronchial tube to ensure good quality product delivered to the market.

2. Scope:

This inspection procedure is applicable for endobronchial tube manufacturing by Saudi Mais Co, for medical products.

3. Definition:

3.1 Endobronchial Tube:

A double-lumen endotracheal tube (also called double-lumen endobronchial tube or DLT) is a type of endotracheal tube which is used in tracheal intubation during thoracic surgery and other medical conditions to achieve selective, one-sided ventilation of either the right or the left lung.

4. Responsibilities:

- 4.1 Quality Controllers are responsible for sampling & testing.
- 4.2 Quality control supervisors are responsible for checking the obtained results.
- 4.3 Machine operators and assembly peoples are responsible to produced good quantity and good quality products.

5. Procedure:

5.1 Receiving Bulk Product:

1.1 Receiving inspection of Bulk Product.

- Ensure that received products are correct as per purchased order.
- Inspect the package condition, must be no damaged and pouch must be with be proper sealed and correct labels.
- Take samples as per AQL general inspection method for visual inspection and AQL special inspection method for physical testing according to the SOP-8-2-6 and make sure that results are same as per required specification.
- Make sure that the product COA available in case if not available submit request to procurement department.

5.1.1 Inspection and test report for Endobronchial tube.

Description	Requirements	Method of check
Packing Condition	Good without any damage	Visual Inspection
Tube Status	Clear and transparent kink resistance / accurate in	Visual / Physical

Prepared by: QA Coordinator
M. Azeem

Approved by: Q.C.M
Mr. Javed Gafoor

Approved by: Q.A.M
Mr. Salman Rashid

Controlled Copy

Saudi Mais Co, for Medical Products Work Instruction

Title

(Chapter, #04)

Inspection Procedure for Endobronchial Tube

Doc. # : WI-8-2-2-06-1

Rev. # : 00

Date : 05.12.2023

Page.# : 2 / 11

	dimension.	
Tube Graduation Scale Status	Should be accurate and bold printed graduation scale, clear and visible	Visual / Physical / Measurement
Tube Dimension	Tube dimension ID, OD and length as per specification	Measurement
X-Ray Line Marking	Clear Straight line, dark blue color,	Visual
Y connector status	Smooth and transparent connector, strong gluing with limbs and tube no pullout, no crack, no damaged	Visual / Physical
Bronchial Limbs (blue) status	Smooth and transparent tube blue color, good fitting with Y connector, strong gluing, no pullout	Visual / Physical
Tracheal Limbs (white) status	Clear and transparent limbs, accurate in dimension, strong gluing with y connector and endo connector.	Visual / Physical
Endo Connector Status	Good fitting with limbs, no flashing, no black spot smooth edge, no crack and broken	Visual / Physical
Inflation tubes tracheal and bronchial Status	Kink Resistance tube, good fitting with tube by strong gluing, no pullout, no blockage and leakage	Visual / Physical
Pilot Balloons and valves	Both tracheal and bronchial pilot balloon should be no leakage, valves should be no crack, no damaged, no leakage, and no blockage.	Visual / Physical
Printing on pilot balloon	Clear printed, visible	Visual
Leak test for pilot balloons	Should be no leakage.	Visual / Physical
Leak test for tracheal and bronchial Cuff	Should be no leakage after inflated.	Visual / Physical
Tip Forming Tracheal status	Smooth cutting, no sharp edges, no blockage	Visual / Physical
Tip Curve Bronchial status	Smooth surface, curved angle, no sharp edges, no blockage, no flashing.	Visual / Physical
Suction Catheter	Clear and transparent catheter smooth surface, no black spots, strong gluing of connector with tube no pullout	Visual / Physical
Swivel Adaptor	Good fitting of Components no damage, no crack, no leakage and free from irregularities	Visual / physical
Stylet status	Smooth surface and flexible, no damaged, no crack	Visual / Physical
Logo	Mais Logo	Visual

5.1.2 Designated Size.

- The designated size in French gauge should be expressed as a whole number the designated size is the tracheal segment OD and be calculated based on the average of the major and minor axis length. The average OD should be within ± 1 mm of the stated size in Fr.

5.1.3 Dimension

- The major or minor axis length across the OD for the tracheal segment should be subjected to the tolerance of ± 0.5 mm.

5.1.4 Main Tube.

- Quality controller is responsible to make sure that the tube should be clear and transparent kink resistance / accurate in dimension as per specification and free from irregularities defects.

Prepared by: QA Coordinator
M. Azeem

Approved by: Q.C.M
Mr. Javeed Gafoor

Approved by: Q.A.M
Mr. Salman Rashid

Controlled Copy

Saudi Mais Co, for Medical Products

Work Instruction

Title (Chapter. #04)
Inspection Procedure for Endobronchial Tube

Doc. #	:	WI-8-2-2-06-1
Rev. #	:	00
Date	:	05.12.2023
Page.#	:	3 / 11

5.1.5 Tracheal Cuff

- Tracheal cuff should be clear and transparent, white Color, accurate in dimension as per specification and should not burst and leak during inflation.
- Check the conformance by inflating the cuff to a pressure of 9.0Kpa (90cm H2O) or to a diameter of 1, 5 times the cuff diameter as determine in ISO5361 Annex B, whichever comes first with the syringe or other inflating devices. Seal the inflating system detach the syringe or other inflating devices.
- Submerge the entire inflating system of the tube in water and observe the bubbles for a period for not less than 10second.

Acceptance Criteria

- No bubble should be noted over the 10 second interval of tracheal cuff.

5.1.6 Bronchial Cuff

- Bronchial cuff should be clear and transparent, blue colour, accurate in dimension as per specification and should not burst and leak during inflation.
- Check the conformance by inflating the cuff to a pressure of 9.0Kpa (90cm H2O) or to a diameter of 1, 5 times the cuff diameter as determine in ISO5361 Annex B, whichever comes first with the syringe or other inflating devices. Seal the inflating system detach the syringe or other inflating devices.
- Submerge the entire inflating system of the tube in water and observe the bubbles for a period for not less than 10second.

Acceptance Criteria

- No bubble should be noted over the 10 second interval of bronchial cuff.

5.1.7 Tracheal and bronchial cuff diameter

- The tracheal and bronchial cuff diameter should be with in $\pm 15\%$ of the marked value when determined in accordance with ISO 5361 ANNEX C, for a non-circular bronchial cuff the cuff resting diameter disclosed should be the diameter through the widest diameter.

5.1.8 Tracheal and bronchial Cuff Herniation

- When tested for cuff herniation according to the method described in ISO 5361 ANNEX E, no part of the inflated cuff should reach the beyond the nearest edge of the patient end of the tracheal segment or the patient end of the bronchial segment only the cuff under test should be inflated.

5.1.9 Tracheal (inflation Tube)

Tracheal connected tube should be kink resistance tube, accurate in dimension as per specification, transparent white color, good fitting with main tube and pilot balloon by strong gluing, and should be free from black spots, holes and blockage.

5.1.10 Bronchial (inflation Tube)

Bronchial connected tube should be kink resistance tube, accurate in dimension as per specification, blue color, good fitting with main tube and pilot balloon by strong gluing, and should be free from black spots, holes and blockage.

Prepared by: QA Coordinator
M. Azeem

Approved by: Q.C.M
Mr. Javeed Gafoor

Approved by: Q.A.M
Mr. Salman Rashid

Controlled Copy

Saudi Mais Co, for Medical Products Work Instruction

Title (Chapter, #04)
Inspection Procedure for Endobronchial Tube

Doc. #	:	WI-8-2-2-06-1
Rev. #	:	00
Date	:	05.12.2023
Page.#	:	4 / 11

5.1.11 Tracheal Pilot Balloon

Clear and transparent white soft balloons with clear printed tracheal, should be strong gluing with inflation tube and valve and free from holes leakage and blockage.

5.1.12 Bronchial Pilot Balloon

Clear and transparent blue color soft balloons with clear printed bronchial, should be strong gluing with inflation tube and valve and free from holes leakage and blockage.

5.1.13 Tracheal and Bronchial Valve

Valve should be strong gluing with pilot balloons, and should be free from leakage after inflated cuff.

5.1.14 Bronchial Limbs (blue)

Should be smooth and transparent tube blue color, good fitting with Y connector by strong gluing, no pullout.

5.1.15 Tracheal Limbs (white)

Should be clear and transparent tracheal limbs, accurate in dimension, good fitting with y connector by strong gluing and proper inserted with endo connector.

5.1.14 Endo Connector Status

Should be good fitting with limbs, proper inserted in tube by tight insertion, no flashing, no black spot / smooth edge, no crack and broken.

5.1.15 4way Connector

Connector should be smooth and transparent connector, strong gluing with limbs and tube no pullout, no crack, no damaged.

5.1.16 X-ray Line

Tube should be with dark blue clear and visible x-ray line.

5.1.17 Swivel 3way Connector

Swivel 3way connectors should be smooth surface good fitting with tubes and should be free from damaged crack.

5.1.18 Swivel Tubes

Clear and transparent both tubes good fitting with three-way tube connector no holes and free from irregularities.

5.1.19 Endoscopic T Connector Cover

Endoscopic connector cover should be flexible good fitting with endoscopic connector damaged and broken.

5.1.20 Endoscopic T Connector

Clear and smooth surface connector good fitting without damaged and crack.

Prepared by: QA Coordinator
M. Azeem

Approved by: Q.C.M
Mr. Javed Gafoor

Approved by: Q.A.M
Mr. Salman Rashid

Controlled Copy

Saudi Mais Co, for Medical Products

Work Instruction

Title

(Chapter. #04)
Inspection Procedure for Endobronchial Tube

Doc. #	:	WI-8-2-06-1
Rev. #	:	00
Date	:	05.12.2023
Page.#	:	5 / 11

5.1.21 Subordinate connector

Subordinate connector rotatable connector good fitting smooth rotation and should be free from damage and crack.

5.1.22 Stylet

Should be aluminum type, smooth surface and flexible, no damaged.

5.1.23 Tip Forming Tracheal status

Should be smooth cutting, round surface, no sharp edges, no blockage.

5.1.24 Tip Curve Bronchial status

Should be smooth surface, curved angle, no sharp edges, no blockage, no flashing.

5.1.25 Endo Connector dimension

The connector of the endobronchial tube complies with the requirement of standard ISO5356-1.

5.1.26 Marking

Marking on the endobronchial tubes should be visible at a distance of $1m \pm 10mm$ in and illuminance of $215 \pm 5lx$, and rubbing the marking by hand without under pressure for 15s, with closed soaked with distilled water the marking should not remove.

5.1.27 Color Coding

The machine end of the bronchial segment, bronchial cuff and its associated pilot balloon should be entirely coloured blue.

The machine end of the Tracheal segment, Tracheal cuff and its associated pilot balloon should be entirely transparent.

5.1.28 Suction Catheters

Visual inspection

When examined by normal vision the catheter and its connectors are free from extraneous matters.

5.1.29 Bonding Strength

The security of the connector forced required to detach any component permanently attached to the shaft as per given table-2 of standard ISO 8836.

Size Outside diameter (mm)	Minimum Force (N)
1.33 up to 2.67	05 N
3.00 up to 4.67	15 N
> 5	20 N

Prepared by: QA Coordinator
M. Azcem

Approved by: Q.C.M
Mr. Javeed Gafoor

Approved by: Q.A.M
Mr. Salman Rashid

Controlled Copy

Saudi Mais Co, for Medical Products Work Instruction

Title (Chapter. #04)
Inspection Procedure for Endobronchial Tube

Doc. # : WI-8-2-2-06-1

Rev. # : 00

Date : 05.12.2023

Page.# : 6 / 11

Visual Inspection

Tube should be transparent and free from irregularities defects.

The tube surface must be smooth, no bumps, no black spot, no burn marks.

The cuffs, two lumens and pilot balloons are imprinted with TRACHEAL (clear) or BROCHIAL (blue Color) for easy identification.

Tube should be with x-ray line and clear line should be clear and visible.

Each component should be strong gluing and no pullout from assembled connectors.

Suction catheter should be clear and smooth surface and strong gluing with connector.

Swivel connector good fitting with components no damage, crack and free from irregularities.

Inprocess Inspection

Quality controllers are responsible to take samples as per AQL general inspection method for visual inspection and AQL special inspection method for physical testing according to the SOP-8-2-6 and collect the samples as per given Table.

Table 1 Sample Size Code Letters

Lot size (Number of ordered products)	Q General Inspection Levels			Q Special Inspection Levels			
	I	II	III	S-1	S-2	S-3	S-4
2 → 8	A	A	B	A	A	A	A
9 → 15	A	B	C	A	A	A	B
16 → 25	B	C	D	A	B	B	C
26 → 50	C	D	E	B	B	C	D
51 → 90	C	E	F	B	B	C	D
91 → 150	D	F	G	B	C	D	E
151 → 280	E	G	H	B	C	D	E
281 → 500	F	H	J	C	C	E	F
501 → 1 200	G	J	K	C	D	E	G
1 201 → 3 200	H	K	L	C	D	F	G
3 201 → 10 000	J	L	M	C	D	F	H
10 001 → 35 000	K	M	N	D	E	G	J
35 001 → 150 000	L	N	P	D	E	G	J
150 001 → 500 000	M	P	Q	D	E	H	K
500 001 → 10000000	N	Q	R	D	E	H	K

Prepared by: QA Coordinator
M. Azeem

Approved by: Q.C.M
Mr. Javeed Gafoor

Approved by: Q.A.M
Mr. Salman Rashid

Controlled Copy

Saudi Mais Co, for Medical Products Work Instruction

Title (Chapter: #04)
Inspection Procedure for Endobronchial Tube

Doc. #	:	WI-8-2-2-06-1
Rev. #	:	00
Date	:	05.12.2023
Page.#	:	7 / 11

SINGLE SAMPLING PLANS FOR NORMAL INSPECTION												
Sample Size Code Letter	Sample Size	Acceptable Quality Levels (Normal Inspection)										
		0.065	0.10	0.15	0.25	0.40	0.65	1.0	1.5	2.5	4.0	6.5
		Ac Re	Ac Re	Ac Re	Ac Re	Ac Re	Ac Re	Ac Re	Ac Re	Ac Re	Ac Re	Ac Re
A	2											0 1
B	3											0 1
C	5											0 1
D	8											0 1
E	13											0 1
F	20											0 1
G	32											0 1
H	50											0 1
J	80											0 1
K	125											0 1
L	200											0 1
M	315											0 1
N	500											0 1
P	800											0 1
Q	1250											0 1
R	2000											0 1

↑ Use first sampling plan above arrow, if sample size equals or exceeds lot or batch size, do 100 percent inspection.
↓ Use first sampling plan below arrow AC : Acceptance number Re : Rejection number

1 Dimension

Quality controllers are responsible to make sure that each components dimension should be accurate as per specification and major or minor axis length across the OD for the tracheal segment should be subjected to the tolerance of $\pm 0.5\text{mm}$.

Main Tube.

Quality controllers are responsible to make sure that the tube should be Clear and transparent kink resistance / accurate in dimension as per specification and free from irregularities defects.

7.3 Tracheal Cuff

Tracheal Cuff should be clear and transparent, white Color, accurate in dimension as per specification and should not burst and leak during inflation.

7.4 Tracheal Cuff Leakage Test

Check the conformance by inflating the cuff to a pressure of 9.0Kpa (90cm H₂O) or to a diameter of 1, 5 times the cuff diameter as determine in ISO5361 Annex B, whichever comes first with the syringe or other inflating devices. Seal the inflating system detach the syringe or other inflating devices. Submerge the entire inflating system of the tube in water and observe the buttlles for a period for not less than 10second.

Acceptance Criteria

No bubble should be noted over the 10 second interval of tracheal cuff.

Prepared by: QA Coordinator
M. Azeem

Approved by: Q.C.M
Mr. Javed Gafoor

Approved by: Q.A.M
Mr. Salman Hashid

Controlled Copy

Saudi Mais Co, for Medical Products Work Instruction

Title (Chapter. #04)
Inspection Procedure for Endobronchial Tube

Doc. #	:	WI-8-2-2-06-1
Rev. #	:	00
Date	:	05.12.2023
Page.#	:	8 / 11

Bronchial Cuff

Bronchial Cuff should be clear and transparent, blue color, accurate in dimension as per specification and should not burst and leak during inflation.

Bronchial Cuff Leakage Test

Check the conformance by inflating the cuff to a pressure of 9.0Kpa (90cm H₂O) or to a diameter of 1, 5 times the cuff diameter as determine in ISO5361 Annex B, whichever comes first with the syringe or other inflating devices. Seal the inflating system detach the syringe or other inflating devices. Submerge the entire inflating system of the tube in water and observe the buttles for a period for not less than 10second.

Acceptance Criteria

No bubble should be noted over the 10 second interval of bronchial cuff.

7.7 Tracheal and bronchial cuff diameter

The tracheal and bronchial cuff diameter should be with in $\pm 15\%$ of the marked value when determined in accordance with ISO 5361 ANNEX C, for a non-circular bronchial cuff the cuff resting diameter disclosed should be the diameter through the widest diameter.

7.8 Tracheal and bronchial Cuff Herniation

When tested for cuff herniation according to the method described in ISO 5361 ANNEX E, no part of the inflated cuff should reach the beyond the nearest edge of the patient end of the tracheal segment or the patient end of the bronchial segment only the cuff under test should be inflated.

7.9 Tracheal (inflation Tube)

Tracheal inflation tube should be kink resistance tube, accurate in dimension as per specification, transparent white color, good fitting with main tube and pilot balloon by strong gluing, and should be free from black spots, holes and blockage.

7.10 Bronchial (inflation Tube)

Bronchial inflation tube should be kink resistance tube, accurate in dimension as per specification, blue color, good fitting with main tube and pilot balloon by strong gluing, and should be free from black spots, holes and blockage.

7.11 Tracheal Pilot Balloon

Clear and transparent white soft balloons with clear printed tracheal, should be strong gluing with inflation tube and valve and free from holes leakage and blockage.

7.12 Bronchial Pilot Balloon

Clear and transparent blue color soft balloons with clear printed bronchial, should be strong gluing with inflation tube and valve and free from holes leakage and blockage.

7.13 Tracheal and Bronchial Valve

Both tracheal and bronchial Valve should be strong gluing with pilot balloons, and should be free from leakage after inflated cuffs.

Prepared by: QA Coordinator
M. Azeem

Approved by: Q.C.M
Mr. Javed Gafoor

Approved by: Q.A.M
Mr. Salman Rashid

Controlled Copy

Saudi M
Quality R

Title: Insp
Accor

Supplier Name

Our P.O.# : -

Received Qty.#

Serial
No

AQL

Saudi Mais Co, for Medical Products
Work Instruction

Title (Chapter. #04)
Inspection Procedure for Endobronchial Tube

Doc. #	:	WI-8-2-2-06-1
Rev. #	:	00
Date	:	05.12.2023
Page.#	:	9 / 11

- 4 **Bronchial Limbs (blue) status**
Bronchial limbs should be smooth and transparent blue color, good fitting with 15mm connector and Y connector, no loose fitting, no pullout, no damage and crack.
- 15 **Tracheal Limbs (white) status**
Tracheal limbs should be clear and transparent accurate in dimension, good fitting with 15mm connector and y connector no loose fitting, no pullout, no damage and crack.
- 16 **Endo Connector Status**
Should be good fitting with limbs, proper inserted in tube by tight insertion, no loose fitting, no flashing, no black spot / smooth edge, no crack and broken.
- 7.17 **Y connector status**
Connector should be smooth and transparent connector, strong gluing with limbs and tube no pullout, no crack, no damaged.
- 7.18 **Main tube X-ray Line**
Tube should be with dark blue clear and visible x-ray line.
- 7.19 **Endoscopic Connector Cover**
Endoscopic connector cover should be flexible good fitting with endoscopic connector damaged and broken.
- 7.20 **Endoscopic Connector**
Clear and smooth surface connector good fitting without damaged and crack.
- 7.21 **Subordinate connector**
Subordinate connector rotatable connector good fitting with subordinate connector, smooth rotation and should be free from damage and crack.
- 7.22 **Stylet**
Should be aluminum type, smooth surface and flexible, no damaged.
- 7.23 **Tip Forming Tracheal status**
Should be smooth cutting, round surface, no sharp edges, no blockage.
- 7.24 **Tip Curve Bronchial status**
Should be smooth surface, curved angle, no sharp edges, no blockage, no flashing.
- 7.25 **Marking**
Marking on the endobronchial tubes should be visible at a distance of 1m±10mm in and illuminance of 215±5lx, and rubbing the marking by hand without under pressure for 15s, with closed soaked with distilled water the marking should not remove.

Prepared by: QA Coordinator
M. Azeem

Approved by: Q.C.M
Mr. Javeed Gafoor

Approved by: Q.A.M
Mr. Salman Rashid

Controlled Copy

Saudi Mais Co, for Medical Products Work Instruction

Title (Chapter. #04)
Inspection Procedure for Endobronchial Tube

Doc. #	:	WI-8-2-06-1
Rev. #	:	00
Date	:	05.12.2023
Page.#	:	10 / 11

26 Color Coding

The machine end of the bronchial segment, bronchial cuff and its associated pilot balloon should be entirely coloured blue.

The machine end of the Tracheal segment, Tracheal cuff and its associated pilot balloon should be entirely transparent.

27 Suction Catheters

Visual inspection

When examined by normal vision the catheter and its connector should be free from extraneous matters.

28 Bonding Strength

The security of the connector forced required to detach any component permanently attached to the shaft as per given table-2 of standard ISO 8836.

Size Outside diameter (mm)	Minimum Force (N)
1.33 up to 2.67	05 N
3.00 up to 4.67	15 N
> 5	20 N

8. Packaging and Labeling:

- Quality controller is responsible to ensure that product packed by proper sealed pouch with correct label information like Product name and code no., reference no., lot no., type, temperature logo, production date, expiry date, and must be stamp on label as approved.
- Every 2 hours QC will take samples from packed quantity base on AQL and record the results in specified Inprocess control form.

9. Final Inspection and Product Release

- Quality controller is responsible to take sample as per AQL general inspection method for visual inspection and AQL special inspection method for physical testing of product and inspect the products as requirements mention in specification and final inspection and release and submit final inspection and release to QCM or QAM for approval.
- After sterilization make sure that each carton must be stamp with chamber number.
- Ensure that the packaging carton must be stamped by packaging persons and quantity of products in packaging carton is same as required.

Prepared by: QA Coordinator
M. Azeem

Approved by: Q.C.M
Mr. Javeed Gafoor

Approved by: Q.A.M
Mr. Salman Rashid

Controlled Copy

Saudi Mais Co, for Medical Products
Work Instruction

Title (Chapter. #04)
Inspection Procedure for Endobronchial Tube

Doc. #	:	W1-8-2-2-06-1
Rev. #	:	00
Date	:	05.12.2023
Page.#	:	11 / 11

10. Reference.

- Inspection and test report for Endobronchial tube : F-823-T98
- Inprocess control for Endobronchial Tube : F-823-i88
- Packaging Inprocess control for Endobronchial Tube : F-823-i4-2
- Final inspection and Release for Endobronchial Tube : F-823-F85
- ISO16628 Anesthetic and respiratory equipment — Tracheobronchial tubes.
- ISO 5361 Anesthetic and respiratory equipment Tracheal tube and connectors
- ISO 8836 suction catheter for use in respiratory tract

12. Modification / Document Change History

Date	Changes History	Rev
05.12.2023	Inspection procedure 1 st issue	00

Prepared by: QA Coordinator
M. Azeem

Approved by: Q.C.M
Mr. Javed Gafoor

Approved by: Q.A.M
Mr. Salman Rashid

Controlled Copy

Saudi
Qual
Title:

Supplier
Our P.O.
Received

Serial
No

AQL